

Day 1 – OOPs (Patient Management System)

Goal: Create patient objects and manage patient records.

Task:

Create a `Patient` class.

Attributes:

- patient_id
- patient_name
- age
- disease
- city

Methods:

- display_patient()
- is_senior_citizen()

Sample Data

Use css

Expected Output

```
text
Senior Citizens:
John
David
```

```
# Day 2 – List Comprehension
```

```
Use Day 1 patient objects.
```

```
Task:
```

```
Extract all diabetes patients.
Extract patients from Chennai.
Create a list of senior citizens.
Create patient summary strings.
```

```
Examples
```

```
["John – Diabetes",
 "Sara – Diabetes"]
```

Bonus

```
["John (65)",  
 "David (70)"]
```

Day 3 – NumPy & Pandas

Create DataFrame from Day 1 data.

Task 1

Calculate:

Average age

Maximum age

Minimum age

Task 2

Disease-wise patient count.

Expected

text

Diabetes 2

Asthma 2

Hypertension 1

Task 3

Add column:

risk_category

Rules:

text

Age >= 60 -> High

Age >= 40 -> Medium

Else Low

Expected DataFrame

Patient	Age	Risk
-----	---	-----
John	65	High
Mary	45	Medium

Day 4 – Logging

Continue Day 3.

Create a script that:

1. Reads patient CSV.
2. Performs validations:

Age cannot be negative

Disease cannot be empty

3. Logs:

INFO

WARNING

ERROR

Sample Logs

text

2026-06-18 INFO Loading patient data

2026-06-18 INFO Total records: 5

2026-06-18 WARNING Missing disease for P006

2026-06-18 ERROR Invalid age for P007

Output Files

text

patients.csv

healthcare.log

Day 5 – PySpark + Unit Testing

Continue same healthcare project.

PySpark Task

Read patient CSV.

Create transformations:

1. Count patients by disease.

```
groupBy("disease").count()
```

2. Count patients by city.

```
groupBy("city").count()
```

3. Count high-risk patients.

```
age >= 60
```

4. Create risk_category column.

High

Medium

Low

5. Save output as Parquet.

Spark Unit Testing

Write test cases for:

Test 1

Input:

age = 65

Expected:

High

Test 2

Input:

age = 45

Expected:

Medium

Test 3

Input:

age = 25

Expected:

Low

Test 4

Verify patient count.

Expected:

5

Test 5

Verify Diabetes patient count.

Expected:

2

Final Deliverable

By the end of Day 5, students will have:

text

healthcare_project/

— patient.py	(OOP)
— patient_tasks.py	(List Comprehension)
— analytics.py	(NumPy + Pandas)
— validator.py	(Logging)
— patients.csv	
— spark_job.py	(PySpark)
— test_spark_job.py	(Unit Testing)
— output.parquet	